

원본 데이터셋에 시계열 데이터가 없어 시간/기간 관련 분석은 제외하였음

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In [ ]: # 라이브러리 불러오기, 데이터 전처리 및 중복 제품 제거
# 최대 할인 상품 목록 중 동일 상품명을 제거 후, 평점 4.0 이상 조건 추가로 가격 혜택과 소비자 만족도를 같이

import pandas as pd
import numpy as np
from pandasql import sqldf

mysql = lambda q: sqldf(q, globals())

# 데이터 불러오기
am = pd.read_csv(r"D:\011.DA_Projects\02.DA_PJT_1\64.SQL\12.Quest\amazon1.csv")

# 원본 보존 후 복사본 생성
am_clean = am.copy()

# 대분류 카테고리 추출
am_clean["main_category"] = am_clean["category"].str.split("|").str[0]

# rating 숫자형 변환
am_clean["rating_num"] = pd.to_numeric(am_clean["rating"], errors="coerce")

# rating_count 숫자형 변환
am_clean["rating_count_num"] = (
    am_clean["rating_count"]
    .astype(str)
    .str.replace(",", "", regex=False)
)
am_clean["rating_count_num"] = pd.to_numeric(am_clean["rating_count_num"], errors="coerce")

# product_id 기준 중복 제거
am_clean = (
    am_clean.sort_values(
        ["product_id", "rating_count_num", "rating_num"],
        ascending=[True, False, False]
    )
    .drop_duplicates(subset=["product_id"], keep="first")
)

am_clean.head()
```

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In [93]: # 전처리. 원본 파일 기준으로 분석용 컬럼 생성

import pandas as pd
import numpy as np
from pandasql import sqldf

mysql = lambda q: sqldf(q, globals())

am = pd.read_csv(r"D:\011.DA_Projects\02.DA_PJT_1\64.SQL\12.Quest\amazon1.csv")

am_raw = am.copy()

am_raw["main_category"] = am_raw["category"].astype(str).str.split("|").str[0]
am_raw["rating_num"] = pd.to_numeric(am_raw["rating"], errors="coerce")

am_raw["rating_count_num"] = (
    am_raw["rating_count"]
    .astype(str)
    .str.replace(",", "", regex=False)
)
am_raw["rating_count_num"] = pd.to_numeric(am_raw["rating_count_num"], errors="coerce")

am_raw["discount_percentage_num"] = (
    am_raw["discount_percentage"]
    .astype(str)
    .str.replace("%", "", regex=False)
)
am_raw["discount_percentage_num"] = pd.to_numeric(
```

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    am_raw["discount_percentage_num"],
    errors="coerce"
)

am_top = am_raw.copy()

am_top = am_top.sort_values(
    ["main_category", "rating_num", "rating_count_num"],
    ascending=[True, False, False]
)

am_top["top_rank_in_category"] = (
    am_top.groupby("main_category").cumcount() + 1
)

am_discount_unique = (
    am_raw
    .sort_values(
        ["discount_percentage_num", "rating_num", "rating_count_num"],
        ascending=[False, False, False]
    )
    .drop_duplicates(subset=["product_name"], keep="first")
)

```

In []: # 추천 시스템 1.

1-1) 이름 : 카테고리별 고평점 상품 추천 시스템

1-2) 테마 : 카테고리별 고평점 상품을 추천하는 시스템

1-3) 구현 로직 : 카테고리별 고평점 및 고평점 갯수 상위 5개 제품으로 한정

1-4) 결과 : 상위 5개 제품 나열 및 5개 미만의 카테고리도 있음

카테고리별 고평점 상품 상위 5개 추출 결과 소비자 만족도가 높은 대표 상품들로 품질 중심 추천에

```

mysql("""
SELECT
    main_category,
    product_name,
    rating_num,
    rating_count_num
FROM am_top
WHERE top_rank_in_category <= 5
ORDER BY
    main_category ASC,
    top_rank_in_category ASC
""")

```

Out[]:

	main_category	product_name	rating_num	rating_count_num
0	Car&Motorbike	Reffair AX30 [MAX] Portable Air Purifier for C...	3.8	1118.0
1	Computers&Accessories	Amazon Basics Wireless Mouse 2.4 GHz Connect...	5.0	23.0
2	Computers&Accessories	Syncwire LTG to USB Cable for Fast Charging Co...	5.0	5.0
3	Computers&Accessories	REDTECH USB-C to Lightning Cable 3.3FT, [Apple...	5.0	NaN
4	Computers&Accessories	Redgear MP35 Speed-Type Gaming Mousepad (Black...	4.6	33434.0
5	Computers&Accessories	Logitech M331 Silent Plus Wireless Mouse, 2.4G...	4.6	12375.0
6	Electronics	Spigen EZ Fit Tempered Glass Screen Protector ...	4.7	7779.0
7	Electronics	Sony Bravia 164 cm (65 inches) 4K Ultra HD Sma...	4.7	5935.0
8	Electronics	Spigen EZ Fit Tempered Glass Screen Protector ...	4.6	26603.0
9	Electronics	Spigen EZ Fit Tempered Glass Screen Protector ...	4.6	6129.0
10	Electronics	10k 8k 4k HDMI Cable, Certified 48Gbps 1ms Ult...	4.6	3664.0
11	Health&PersonalCare	Dr Trust Electronic Kitchen Digital Scale Weig...	4.0	3663.0
12	Home&Kitchen	Swiffer Instant Electric Water Heater Faucet T...	4.8	53803.0
13	Home&Kitchen	Instant Pot Air Fryer, Vortex 2QT, Touch Contr...	4.8	3964.0
14	Home&Kitchen	Oratech Coffee Frother electric, milk frother ...	4.8	28.0
15	Home&Kitchen	Campfire Spring Chef Prolix Instant Portable W...	4.7	2591.0
16	Home&Kitchen	Multifunctional 2 in 1 Electric Egg Boiling St...	4.7	2300.0
17	HomeImprovement	Gizga Essentials Cable Organiser, Cord Managem...	4.5	5985.0
18	HomeImprovement	ESnipe Mart Worldwide Travel Adapter with Buil...	4.0	2581.0
19	MusicalInstruments	Boya ByM1 Auxiliary Omnidirectional Lavalier C...	4.0	68664.0
20	MusicalInstruments	MAONO AU-400 Lavalier Auxiliary Omnidirectiona...	3.8	20218.0
21	OfficeProducts	Classmate Soft Cover 6 Subject Spiral Binding ...	4.5	8618.0
22	OfficeProducts	Casio MJ-12D 150 Steps Check and Correct Deskt...	4.5	8610.0
23	OfficeProducts	Casio FX-991ES Plus-2nd Edition Scientific Cal...	4.5	5760.0
24	OfficeProducts	Classmate Drawing Book - Unruled, 40 Pages, 21...	4.5	4951.0
25	OfficeProducts	Classmate Soft Cover 6 Subject Spiral Binding ...	4.5	4428.0
26	Toys&Games	Faber-Castell Connector Pen Set - Pack of 25 (...)	4.3	15867.0

In []:

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# 추천 시스템 2.
# 2-1) 이름 : 카테고리별 인기상품 추천 시스템
# 2-2) 테마 : 리뷰 수를 기준으로 카테고리별 리뷰 많은 상품을 추천하는 시스템
# 2-3) 구현 로직 : rating_count_sum IS NOT NULL 조건 설정 및 rating_num + rating_count_num 내림차순
# 2-4) 결과 : 상위 5개 제품 나열 및 5개 미만의 카테고리도 있음
#           카테고리별로 리뷰수가 많은 상품 조회 결과 실제 소비자 반응이 활발한 인기 상품들을 확인할 수 있고

mysql("""
SELECT
    main_category,
    product_name,
    rating_num,
    rating_count_num
FROM am_raw
WHERE rating_count_num IS NOT NULL
ORDER BY
    main_category ASC,
    rating_count_num DESC,
    rating_num DESC
""")
```

Out[]:

	main_category	product_name	rating_num	rating_count_num
0	Car&Motorbike	Reffair AX30 [MAX] Portable Air Purifier for C...	3.8	1118.0
1	Computers&Accessories	SanDisk Cruzer Blade 32GB USB Flash Drive	4.3	253105.0
2	Computers&Accessories	SanDisk Ultra Dual 64 GB USB 3.0 OTG Pen Drive...	4.3	189104.0
3	Computers&Accessories	TP-Link Nano USB WiFi Dongle 150Mbps High Gain...	4.2	179692.0
4	Computers&Accessories	TP-Link USB WiFi Adapter for PC(TL-WN725N), N1...	4.2	179691.0
...	...	...	...	...
1458	OfficeProducts	Classmate Long Book - Unruled, 160 Pages, 314 ...	4.5	1674.0
1459	OfficeProducts	Classmate Pulse Spiral Notebook - 240 mm x 180...	4.1	1269.0
1460	OfficeProducts	Classmate Pulse 1 Subject Notebook - 240mm x 1...	4.2	419.0
1461	OfficeProducts	BRUSTRO Copytinta Coloured Craft Paper A4 Size...	4.3	388.0
1462	Toys&Games	Faber-Castell Connector Pen Set - Pack of 25 (...)	4.3	15867.0

1463 rows × 4 columns

In [95]:

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# 추천 시스템 3.
# 3-1) 이름 : 최대 할인율 기반 추천 시스템
# 3-2) 테마 : 할인율을 기준으로 할인율이 높은 상품을 추천하는 시스템
# 3-3) 구현 로직 : 전체 상품 중 할인율이 가장 높은 상품을 조회하는 쿼리 (할인율 우선 정렬 + 평점 + 리뷰수 순)
# 3-4) 결과 : 전체 데이터셋에서 할인율 높은 상품 추출 결과 가격 혜택이 큰 특가 상품들을 확인할 수 있었음,
#          (동일 상품명에 중복되는 경우가 있어 원본 데이터 기준 결과임을 함께 설명 필요)

mysql("""
SELECT
    main_category,
    product_name,
    discount_percentage_num,
    rating_num,
    rating_count_num
FROM am_raw
WHERE discount_percentage_num IS NOT NULL
ORDER BY
    discount_percentage_num DESC,
    rating_num DESC,
    rating_count_num DESC
LIMIT 20
""")
```

Out[95]:

	main_category	product_name	discount_percentage_num	rating_num	rating_count_num
0	Computers&Accessories	rts [2 Pack] Mini USB C Type C Adapter Plug, T...	94	4.3	4426.0
1	Electronics	Fire-Boltt Ninja Call Pro Plus 1.83" Smart Wat...	91	4.2	13937.0
2	Electronics	Fire-Boltt Ninja Call Pro Plus 1.83" Smart Wat...	91	4.2	13937.0
3	Electronics	Fire-Boltt Ninja Call Pro Plus 1.83" Smart Wat...	91	4.2	13937.0
4	Electronics	Fire-Boltt Ninja Call Pro Plus 1.83" Smart Wat...	91	4.2	13937.0
5	Electronics	Fire-Boltt Ninja Call Pro Plus 1.83" Smart Wat...	91	4.2	13937.0
6	Electronics	LAPSTER 12pcs Spiral Cable Protectors for Char...	90	4.4	305.0
7	Computers&Accessories	LAPSTER Spiral Charger Spiral Charger Cable Pr...	90	4.1	8751.0
8	Electronics	Sounce Protective Case Cover Compatible Boat X...	90	4.0	4740.0
9	Computers&Accessories	Sounce Spiral Charger Cable Protector Data Cab...	90	4.0	1396.0
10	Computers&Accessories	Sounce Spiral Charger Cable Protector Data Cab...	90	4.0	1396.0
11	Computers&Accessories	Sounce Fast Phone Charging Cable & Data Sync U...	90	3.9	7928.0
12	Electronics	Silicone Rubber Earbuds Tips, Eartips, Earpads...	90	3.8	594.0
13	Home&Kitchen	beatXP Kitchen Scale Multipurpose Portable Ele...	90	3.7	2031.0
14	Computers&Accessories	pTron Solero M241 2.4A Micro USB Data & Chargi...	89	3.9	1075.0
15	Electronics	Rts™ High Speed 3D Full HD 1080p Support (10 M...	88	4.2	910.0
16	Computers&Accessories	PTron Solero T241 2.4A Type-C Data & Charging ...	88	3.9	24871.0
17	Electronics	Sounce Gold Plated 3.5 mm Headphone Splitter f...	88	3.9	6491.0
18	Electronics	Sounce Gold Plated 3.5 mm Headphone Splitter f...	88	3.9	6491.0
19	Computers&Accessories	pTron Solero T241 2.4A Type-C Data & Charging ...	88	3.9	1075.0

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# 추천 시스템 4.
# 4-1) 이름 : 가성비 추천 시스템 (고할인율 + 평점 > 4.0)
# 4-2) 테마 : 할인율을 기준으로 할인율이 높고 평점 > 4.0 상품 추천 시스템
# 4-3) 구현 로직 : 중복 상품명 제거 후, 평점 4.0 이상 조건을 넣어 만든 가성비 추천 쿼리
# 4-4) 결과 : 고할인율 제품 중 고사용자 만족도와 가격 메리트를 함께 반영한 추천 시스템입니다

mysql("""
SELECT
    main_category,
    product_name,
    discount_percentage_num,
    rating_num,
    rating_count_num
```

```

FROM am_discount_unique
WHERE discount_percentage_num IS NOT NULL
AND rating_num >= 4.0
ORDER BY
    discount_percentage_num DESC,
    rating_num DESC,
    rating_count_num DESC
LIMIT 20
""")

```

Out[]:

	main_category	product_name	discount_percentage_num	rating_num	rating_count_num
0	Computers&Accessories	rts [2 Pack] Mini USB C Type C Adapter Plug, T...	94	4.3	4426.0
1	Electronics	Fire-Boltt Ninja Call Pro Plus 1.83" Smart Wat...	91	4.2	13937.0
2	Electronics	LAPSTER 12pcs Spiral Cable Protectors for Char...	90	4.4	305.0
3	Computers&Accessories	LAPSTER Spiral Charger Spiral Charger Cable Pr...	90	4.1	8751.0
4	Electronics	Sounce Protective Case Cover Compatible Boat X...	90	4.0	4740.0
5	Computers&Accessories	Sounce Spiral Charger Cable Protector Data Cab...	90	4.0	1396.0
6	Electronics	Rts™ High Speed 3D Full HD 1080p Support (10 M...	88	4.2	910.0
7	Computers&Accessories	STRIFF Mpad Mouse Mat 230X190X3mm Gaming Mouse...	87	4.2	491.0
8	Computers&Accessories	Lapster 1.5 mtr USB 2.0 Type A Male to USB A M...	86	4.0	1313.0
9	Electronics	Elv Mobile Phone Mount Tabletop Holder for Pho...	85	4.3	2351.0
10	Electronics	Fire-Boltt Ninja 3 Smartwatch Full Touch 1.69 ...	85	4.2	22638.0
11	Computers&Accessories	Lapster 5 pin mini usb cable, usb b cable,came...	85	4.0	1313.0
12	Electronics	STRIFF 12 Pieces Highly Flexible Silicone Micr...	84	4.2	1949.0
13	Electronics	Posh 1.5 Meter High Speed Gold Plated HDMI Mal...	83	4.3	1237.0
14	Computers&Accessories	Callas Multipurpose Foldable Laptop Table with...	83	4.0	20457.0
15	Electronics	Electvision Remote Control Compatible with Kod...	83	4.0	343.0
16	Electronics	ELV Aluminum Adjustable Mobile Phone Foldable ...	82	4.5	28978.0
17	Electronics	Caldipree Silicone Case Cover Compatible for 2...	82	4.3	407.0
18	Electronics	STRIFF Wall Mount Phone Holder Wall Mount with...	82	4.1	9340.0
19	Electronics	pTron Volta Dual Port 12W Smart USB Charger Ad...	82	4.0	3197.0

In []: # 추천 시스템 5.
5-1) 이름 : 카테고리별 프리미엄 추천 시스템(평점 4.5 이상 중심)
5-2) 테마 : 매우 높은 평점을 가진 프리미엄 상품을 추천하는 시스템입니다.
5-3) 구현 로직 : 카테고리별 상위 상품 중 평점 > 4.5 상품만 추출하여 프리미엄 상품군 구성
5-4) 결과 : 고품질 상품을 확인할 수 있었으며, 만족도 중심 추천에 적합한 결과를 도출

```
mysql("""
SELECT
    main_category,
    product_name,
    rating_num,
    rating_count_num
FROM am_top
WHERE top_rank_in_category <= 5
    AND rating_num >= 4.5
ORDER BY
    main_category ASC,
    top_rank_in_category ASC
""")
```

Out[]:

	main_category	product_name	rating_num	rating_count_num
0	Computers&Accessories	Amazon Basics Wireless Mouse 2.4 GHz Connect...	5.0	23.0
1	Computers&Accessories	Synwire LTG to USB Cable for Fast Charging Co...	5.0	5.0
2	Computers&Accessories	REDTECH USB-C to Lightning Cable 3.3FT, [Apple...	5.0	NaN
3	Computers&Accessories	Redgear MP35 Speed-Type Gaming Mousepad (Black...	4.6	33434.0
4	Computers&Accessories	Logitech M331 Silent Plus Wireless Mouse, 2.4G...	4.6	12375.0
5	Electronics	Spigen EZ Fit Tempered Glass Screen Protector ...	4.7	7779.0
6	Electronics	Sony Bravia 164 cm (65 inches) 4K Ultra HD Sma...	4.7	5935.0
7	Electronics	Spigen EZ Fit Tempered Glass Screen Protector ...	4.6	26603.0
8	Electronics	Spigen EZ Fit Tempered Glass Screen Protector ...	4.6	6129.0
9	Electronics	10k 8k 4k HDMI Cable, Certified 48Gbps 1ms Ult...	4.6	3664.0
10	Home&Kitchen	Swiffer Instant Electric Water Heater Faucet T...	4.8	53803.0
11	Home&Kitchen	Instant Pot Air Fryer, Vortex 2QT, Touch Contr...	4.8	3964.0
12	Home&Kitchen	Oratech Coffee Frother electric, milk frother ...	4.8	28.0
13	Home&Kitchen	Campfire Spring Chef Prolix Instant Portable W...	4.7	2591.0
14	Home&Kitchen	Multifunctional 2 in 1 Electric Egg Boiling St...	4.7	2300.0
15	HomeImprovement	Gizga Essentials Cable Organiser, Cord Managem...	4.5	5985.0
16	OfficeProducts	Classmate Soft Cover 6 Subject Spiral Binding ...	4.5	8618.0
17	OfficeProducts	Casio MJ-12D 150 Steps Check and Correct Deskt...	4.5	8610.0
18	OfficeProducts	Casio FX-991ES Plus-2nd Edition Scientific Cal...	4.5	5760.0
19	OfficeProducts	Classmate Drawing Book - Unruled, 40 Pages, 21...	4.5	4951.0
20	OfficeProducts	Classmate Soft Cover 6 Subject Spiral Binding ...	4.5	4428.0